

Demand Is Not Weather

A large share of what a supply chain calls demand is the echo of its own decisions. Evidence from six engagements on what changes when the decision, and not the forecast, becomes the object.

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EXECUTIVE CONCLUSION

Forecasting treats demand as weather: something that arrives from outside, indifferent to the firm, improvable only by predicting it more accurately. A large share of demand does not behave that way.

What sat on the shelf, at what price, quoted at what lead time, allocated to which account, promoted in which week. Each of those produces demand as much as it reveals it. A company forecasting those effects is predicting the consequences of choices it already made, then holding inventory to absorb its own surprise. The instruments layered on top of that arrangement, accuracy targets and safety stocks and service levels and ABC classes, are percentages standing where a price belongs.

01	02	03
Separate the demand you receive from the demand you cause. Only the first is a forecasting problem.	Forecast distributions, lead times included. A mean with a factor bolted onto it discards the shape that generates the cost.	Rank candidate decisions in dollars against one pool of capital. Percentages cannot be compared or added up.

On the evidence

Six engagements are drawn on here, from distribution, discrete manufacturing, retail, consumer goods, and field service. Names are gone, two are welded together from more than one client, and any number specific enough to identify a company has been changed. Six engagements make a caseload, not a sample, and consulting work skews toward companies that already suspect something is off.

The argument owes a debt it should state plainly. Lokad, and Joannes Vermorel in particular, have made the case for probabilistic forecasting, economic drivers, and dollars rather than percentages as the unit of supply chain measurement. Warren Powell’s sequential decision analytics supplies the vocabulary for policies. What follows is what those ideas look like from inside six engagements.

1. The part of demand you are causing

A forecast is an estimate of something the firm does not control. That definition is worth taking seriously, because it disqualifies a good deal of what gets forecast.

Consider a manufacturer deciding which accessories to attach to a vehicle line. Attach rates are routinely forecast, tracked against accuracy targets, and reviewed monthly. Yet the attach rate is largely manufactured: it responds to which accessories are bundled, how they are priced, what the dealer is incentivized to present, and whether the part is available at all. The number being predicted is the output of a decision the firm makes and can change. Predicting it is a strange use of effort when setting it deliberately is available.

Availability is the sharpest case, because the feedback runs through the data itself. A stockout suppresses recorded sales. The suppressed record enters history as demand. History trains the forecast down. The smaller forecast justifies a smaller buy, and the smaller buy produces the next stockout. The series converges,

and what it converges to is a description of the firm’s own past caution.

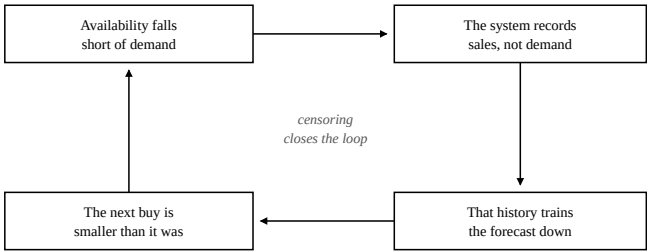


Figure 1. Demand you caused, recorded as demand you observed. A stockout destroys the evidence of the demand it destroyed. Fit a model to the surviving record and it learns the firm’s own caution, then recommends more of it.

A serviceable test separates the two kinds. Would this number have come out differently if we had decided differently? Weather fails that test, which is what makes it weather. Attach rates, promotional lift, quoted lead times, and observed sales during a shortage all pass it comfortably, and none of them belong in a forecast accuracy review as though the firm were a bystander.

FIELD LESSON

Before funding an accuracy program, sort the series into what arrives and what the firm produces. Accuracy work on the second kind is effort spent predicting your own past decisions.

2. Instruments standing where a price belongs

Suppose the sorting is done and genuine uncertainty remains. It has to be absorbed somewhere: in inventory, in capacity, in time, in cash, or by the customer. Choosing among those requires

prices. Most supply chains do not have them, and use instruments instead.

Four weeks of cover

A durables importer asked for a better demand forecast. Safety stock stood at four weeks across effectively the whole catalog. Asked where four weeks came from, the planning manager said it was what they went to after the port thing. Three of the four people who had been in that room had left the company.

The textbook expression for safety stock under demand and lead time variability:

$$SS = z \sqrt{\bar{L} \sigma_D^2 + \bar{D}^2 \sigma_L^2}$$

It asks for two things. The first is that demand and lead time are both roughly normal. The second is that this item can be provisioned on its own, with capital that belongs to it.

Neither held. Their lead times were not bell shaped in any sense: most receipts cleared in about the same few days, and a subset of tariff codes went sideways for weeks. A spike with a long right tail is not a distribution a z multiplier can summarize, and the number the formula returned described a world the data did not contain. Splitting the recorded lead time into supplier confirmation, production, ocean transit, customs, and drayage into the DC located the two tails: customs on that subset of codes, and appointment scheduling at their own building. They were financing inventory against a calendar problem forty feet from the receiving door.

Lead time deserves the same treatment as demand, forecast as a distribution over days and composed with the demand distribution, rather than compressed to an average and marked up by a service factor. The second assumption is the more expensive one, and section 3 returns to it.

A 98% fill rate

A specialty retailer held a 98% fill rate across the assortment. The number appeared in the supply chain scorecard, in vendor agreements, and in a board deck.

For a single-period stocking decision the optimal service level is the critical ratio:

$$q^* = C_u / (C_u + C_o)$$

with C_u the cost of one unit short and C_o the cost of one unit long. Set a 98% fill rate and you have committed to a ratio of 49 to 1. One unit short costs forty-nine times one unit over.

That may well be true for the item anchoring a category and pulling attachments. It is rarely true for the item with a near-identical substitute on the same shelf, where a stockout mostly moves the sale and the honest ratio was closer to three. For a seasonal item eight weeks from markdown, C_o climbed every week while the target held still. The 98% had arrived from a director's previous employer, in a different category, roughly a decade earlier.

The tempting repair is to cut the catalog into classes and give each its own target. That reproduces the original problem at higher

resolution, since a class is still a percentage standing in for a price, and the boundaries end up argued rather than computed.

What the retailer did instead was drop the target. For each item, the expected return in dollars from holding one more unit was computed against the demand and lead time distributions, declining as units were added because the second unit earns less than the first. Every candidate unit in the assortment then entered one ranked list, and the list was bought down until the capital allocated to inventory ran out. The hero item and the commodity item stopped belonging to different classes and began competing directly, unit against unit.

3. One pool of capital, one list

The ranked list is the part that carries. An item-by-item policy, whatever generates it, quietly assumes each item can be funded on its own terms. No warehouse works that way. The items compete for the same cash, the same dock, the same shelf, and the same working capital line, so the only defensible comparison is between one candidate unit and another.

Two of the engagements showed what happens to the terms in that comparison when nobody prices them.

Plan churn

A tier-one automotive supplier ran weekly MRP regeneration and got materially different plans from nearly identical inputs. Released quantities moved, sequencing moved, supplier releases followed. The planners had built the usual defenses: maintained spreadsheets, a private set of rules, and the habit of running the official plan through personal judgment before anything left the building. One of them called his workbook the real plan, and he meant it literally.

The scoped project was forecast improvement, on the theory that a steadier forecast would steady the plan. The planning run re-optimized from scratch each cycle with no term for the cost of changing a commitment already made. Changing a plan costs setup and changeover, credibility with suppliers who staffed to the last release, labor scheduling, and the standing belief downstream that a release means something. The objective priced change at zero, so the solver spent it chasing small theoretical gains off noise.

$$\min \sum_t c_t x_t + \lambda \sum_t |x_t - \tilde{x}_t|$$

Here $\tilde{x}$ is the plan already communicated and λ prices the change. λ has a dollar value, and the planners produced a defensible one in about an hour, because they had been paying it for years. Churn dropped against a small and now-visible loss in theoretical optimality, a trade that had been running the whole time in the wrong direction with nobody's signature on it.

Expediting

An industrial distributor tracked expedited freight as an unfavorable variance and reviewed branch managers on it monthly. The branches responded rationally: expedites were split below reporting thresholds, booked into other freight categories,

arranged straight with carriers outside the TMS, or pre-empted by holding extra stock nobody had authorized. The variance line improved every quarter while the freight bill held steady.

Expediting is recourse, the second-stage action in a problem where you commit, observe, then correct. Under genuine uncertainty some recourse is economically correct, and a distributor with near-zero expedite spend is over-buffered somewhere else at higher cost, in inventory that looks healthy until the day it is written down. Budgeting expedite at an explicit unit price made the spend legitimate, which made it recorded, which turned it into a daily map of where the buffers were wrong.

4. The system reporting its own defects

Both remaining engagements involved organizations that were already generating the diagnosis and discarding it.

An accuracy metric with nowhere to go

A telecom field services group forecast spare parts and was managed on MAPE. Much of the catalog was intermittent: parts moving a handful of times a year, at a particular site, with long runs of zeros. MAPE is undefined at zero and unstable near it, so the team reported at region and month, where the series aggregate into a stable denominator and the metric looks respectable. The stocking decision was made at site and week.

Nobody there would have called it gaming. They were responding correctly to a measurement with no honest room to improve at the granularity where the decision lived. The intermittent catalog made the defect visible without being its cause. A percentage says nothing about what an error costs, and a forecast can improve on

MAPE while the decisions taken from it get worse, when the gain lands on cheap items and the loss lands on expensive ones. Their real question was denominated in dollars: the cost of a truck roll, against a part idle on a shelf for two years, against what the asset does to a customer while it is down.

The override log

A consumer goods manufacturer had a working allocation optimizer and an override rate high enough that leadership had scoped the next phase as adoption, meaning training, a redesigned interface, and a communications plan. The override was applied in a downstream screen and kept nowhere. Once the plan published, the recommendation it replaced was gone.

An override rate is a defect rate. Each one records that the recipe produced a number somebody with local knowledge judged wrong, and thousands per week were being reported into a void. Capturing them with a reason code took a few days. Within a month the codes had collapsed into five buckets, three of which carried most of the volume: named-account allocation commitments made by sales, promotional protection for accounts in a live program, and a minimum-shipment convention with two customers whose receiving docks could not take partial pallets. All of it real, all of it contractual, none of it in the model.

Encoding those three retired most of the override traffic, which was the point. A planner correcting the same class of error every morning is unpaid labor covering a gap in the numerical recipe, and the gap costs less to close than to staff. An adoption program that makes overrides easier to perform and better tracked, while leaving the cause in place, industrialises the defect.

THE SIX ENGAGEMENTS

Setting	What they asked for	What was binding	What changed
Durables importer	A better demand forecast	A lead time with a fat tail, described by a formula that assumes it has none	Lead time forecast as a distribution per stage, composed with demand
Specialty retailer	A 98% fill rate, more cheaply	One percentage standing in for several hundred distinct prices	Target dropped; candidate units ranked in dollars and bought down to a capital limit
Automotive tier one	A steadier forecast, to steady the plan	Zero price on changing a release already communicated	Change penalty in the objective, with λ estimated by the planners
Industrial distributor	Less expedite spend	Recourse booked as failure, so it moved off the books instead of down	Expedite priced and budgeted; spend logged by lane and item
Telecom field services	Better spare parts accuracy	A percentage metric that could not move honestly at the decision's granularity	Stocking measured in dollars: truck roll against holding against downtime
Consumer goods maker	Better planner adoption	Contractual commitments that were real, binding, and absent from the model	Overrides read as defect reports; the three main causes encoded

5. What this implies for sequencing

Most supply chain analytics programs run in technology order: data, forecast, optimization, dashboard, adoption. Every step is defensible, which is part of why the sequence survives contact with a steering committee. The order that follows from the argument above is different.

1. **Sort the demand.** What arrives, against what the firm produces through price, assortment, allocation, and availability.
2. **Write the economic drivers down in currency.** Shortage, excess, delay, plan change, expediting, and the customer's patience.
3. **Forecast distributions, demand and lead time both,** and compose them rather than layering a factor on a mean.

4. **Rank every feasible decision in dollars** against one pool of capital, and truncate the list where the capital stops.
5. Automate the mundane decisions, so attention goes to the recipe instead of the queue.
6. **Put a name against the recipe.** One person owns the numbers end to end and answers for them.

Forecasting keeps a place in this. It enters after the sorting, applied to the part of demand the firm genuinely does not control, and it is judged by the dollars the resulting decisions earn. On two of the six engagements that produced a real forecasting project, better specified for having been asked in fifth position rather than first.

6. Three objections worth answering

Most demand really is exogenous

Often true, and the claim here is narrower than the objection assumes. Nobody controls whether a customer's plant burns down or a competitor exits a category. The point of the sorting step in section 5 is to find the share that the firm does control, which in the engagements above was consistently larger than anyone had assumed, and which had been filed as a forecasting problem because that is where series go by default. A company that has never done the sort has no basis for asserting the share is small.

Ranking every unit needs data we do not have

The compute stopped being the constraint some years ago. Ranking a few million candidate units against a demand and lead time distribution is an overnight job on ordinary hardware. What is usually missing is the prices, and the reason they are missing is that no instrument in the standard toolkit ever asked for them. A service level target can be set by a committee in an afternoon; a cost of shortage requires somebody to take a position in writing.

Precision is also less demanding than it looks. A ranked list built on rough economics puts the right items near the top and the wrong items near the bottom, which is the whole job. A percentage target applied uniformly gets the ordering wrong no matter how precisely it is measured.

Our service level is contractual

Then it is a constraint and should be modeled as one, priced against the penalty in the contract and applied to the accounts and items the contract actually covers. A contractual 98% on two customers is a different object from a 98% policy applied across an entire assortment because two customers negotiated it. In three of the six engagements, a target described as contractual turned out on inspection to cover a minority of volume and to have been generalized outward by convention.

THE CORE TAKEAWAY

Work out which part of demand you are causing before you spend anything on predicting it, then price what remains and rank the decisions in dollars. Accuracy targets, safety stocks, and service levels are what a supply chain uses in place of that arithmetic.

ABOUT BIT BROS

Bit Bros is a boutique applied economics firm working on supply chain and decision systems, at the intersection of optimization, simulation, forecasting, and AI. Founder experience includes Amazon, Toyota, Gurobi, and AT&T. We work on the operating mechanism around the model: how decisions get specified, priced, ranked, automated, and owned.

RELATED READING

Lokad's quantitative supply chain writing, particularly Joannes Vermorel on probabilistic forecasting, prioritized ordering, and measuring error in dollars. Warren Powell on sequential decision analytics. *The Decision Factory* (DeJans and Elam, 2026) puts the argument in narrative form.

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